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Document Approval:

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Leinani Roylo
Leinani Roylo (Dec 17, 2024 10:32 PST)

• Content/Format • Content/ Format, Requirement ID's/Source ID's, Verification Planning, SME Input

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• Scope Alignment, Stakeholder/ SME Engagement, as appl

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Eliazar Ortiz
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• Articulation, Meets Higher-Level Expectations, Stakeholder/SME Engagement

See [Document Review/Approval Matrix of Responsibilities](#)

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Revision History

Revision	Date Released	Description of Change
R1	12/17/2024	Added more information, such as links to drawings, roles and responsibilities, and required sign offs of steps during assembly. Tailored the Traveler for PF2L2 specifcally.
R0	11/30/2022	Original Release.

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1 Purpose

This Plans and Procedures Document is a Traveler that describes the assembly process for the LCLS-II-HE Wavefront Sensor (WFS) as specified in the table below. The final product once completing this Traveler will be a Ready for Installation (RFI) device.

Top Level Assembly Part Number	DSG-000043770, HE Wave Front Sensor Assembly
Asset Name	PF2L2
Vacuum Level	Level 3
Assembly Folder	PF2L2 - LDVC0004001 - 0743
ServiceNow ID	LDVC0004001

2 Acronyms

RGA	Residual Gas Analyzer
ASSY	Assembly
QA	Quality Assurance
NIL	No Indicated Leak
RTD	Resistance Temperature Detector
UHV	Ultra-High Vacuum
SOW	Statement of Work

3 Roles and Responsibilities

Role	Name	Contact Information
Responsible Technician(s)		
Primary Responsible Mechanical Engineer	Leinani Roylo	lroylo@slac.stanford.edu ext. 2834
Secondary Resp. ME	Josh Kirks	jkirks@slac.stanford.edu ext. 3520
Responsible ECS Engineer	Divya Kameswaran	divyak@slac.stanford.edu ext. 2747

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4 References

Document No.	Revision No.	Document Title
L2SI-PR-0134-R0	R0	L2SI Wavefront Sensor Target (WFS) System Requirements
DSG-000043770	R01	HE WAVE FRONT SENSOR ASSY
LCL5004-001164	R00	VACUUM CHAMBR ASSY 6 PORT
LCL5002-003852	R04	WFS STAGE ASSEMBLY
DSG-000044220	R01	HE WFS TOWER ASSY
LCL5004-004358	R00	STAGE TOWER ASSY
LCL5004-003026	R00	BASE PLATE ASSY
LCL5004-004413	R02	CABLE MOUNT BRACKET
LCL5004-004420	R00	DIN RAIL
LCL5004-004692	R00	ELECTRONICS COVER
DSG-000067212	R00	ACTR-MOTOR, 406XR-SZ 34, 100MM TRAVEL
LCL5010-006088	R01	MOTOR COVER
DSG-000043946	R00	HE TARGET MOVER ASSY
LCL5006-005605	R05	LADDER COOLING ASSY
LCL5010-005721	R00	COOLING LOOP CLAMP
LCL5004-001216	R01	6.0 CF FLG BELLOWS ASSY
LCL5010-005817	R00	CLAMP BAR
LCL5010-005627	R00	LADDER NUT BLOCK
DSG-000043817	R01	TARGET ASSEMBLY
DSG-000043789	R01	TARGET FRAME PLATE
DSG-000043818	R01	TARGET BASE PLATE
LCL5006-006848	R00	SENSOR DIAMOND
DSG-000001357	R00	BRACKET, WATER FITTING

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WD-391-701-08	R00	L2HE Wave Front Sensor Wiring Diagram
SLAC-I-030-714-001-06	R000	LCLS Specification and Guidelines for Level 3 Vacuum Qualification
LCLSII-2.4-EN-0726-R1	R1	LCLS-II BC1B-COL1B Warm Region Fastener Torques
SLAC-I-007-12001-006	R000	Clean Room Practices
LCLSII-HE-1.4-ES-1135	R0	LCLS-II-HE Wavefront Sensor Diamond Targets Specification
SLAC-I-030-714-001-03	R003	LCLS Procedure for Vacuum Qualification RGA Scans

5 Requirements During Assembly

- All work done in the clean room, especially when handling UHV components, will be done in accordance with [SLAC-I-007-12001-006 “Clean Room Practices”](#).
- The responsible party shall type their name within this document in the “[Type your name here]” areas and add the corresponding date in the “[Place date here]” areas.
 - Date format: Month/Day/Year (e.g., 01/01/2024)
- All required data shall be entered into the appropriate tables in each section.
- Any file that needs to be attached to this traveler will be uploaded to the designated folder linked in Section 1.
- Labels shall be added to motors connected to the Beckhoff. Labels shall adhere to the following format: *Device_Name-Device-MMS-Motion_Axis*
 - Example: PF2L2-WFS-MMS-Z
- All critical measuring and test equipment used that require calibration shall be documented in Section 7.
- In general, sections shall be performed sequentially. However, the assembly team may use their discretion and communicate with the Responsible Mechanical Engineer when steps can be performed in parallel or in a different ordered sequence.

6 Beam Line Device Names and Serialization

- Each assembled device shall be labeled with the top-level assembly part number (e.g., DSG-000043770) and device name (e.g., PF2L2) as specified in Section 1.
- Create (2X) labels with part number and device name and (2X) Beam Direction Labels.

Beam Direction Examples:

➡ BEAM DIRECTION ➡

⬅ BEAM DIRECTION ⬅

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7 Measuring and Test Equipment

If there are any problems or concerns, contact the Responsible Mechanical Engineer

For all critical measuring and testing equipment, such as the RGA carts used to perform leak checks and provide vacuum qualification, fill in the table below with the following data:

1. Equipment Name and Identifier
2. Confirmation of Calibration
 - a. Provide a short description of how calibration was confirmed.
 - i. Example: RGA was calibrated per Step 3.1 in [SLAC-I-030-714-001-03](#).
3. Tester and Date
 - a. Tester: Person who has used the device and logged the equipment in the table.
 - b. Date: When the tool was used.
4. Notes
 - a. Provide any additional important information.

Equipment Name/ID	Confirmation of Calibration	Tester and Date	Notes
	[How calibration was confirmed]	[Type your name here] [Place date here]	

8 Processing Data (Bake Out, Leak Check, etc.)

Responsible Person(s): Kitting Team and/or Assembly Technician

If there are any problems or concerns, contact the Responsible Mechanical Engineer.

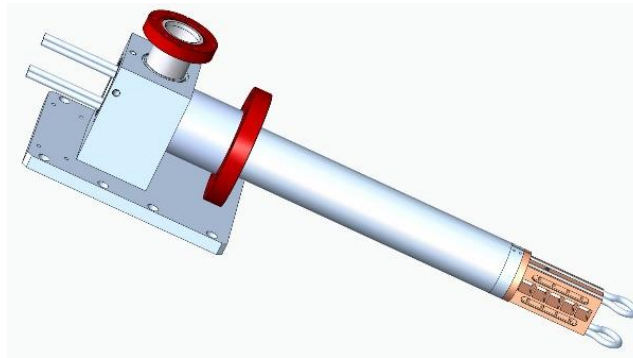
8.1 6-Port Vacuum Chamber ([LCL5004-001164](#)) Data Check

Each vacuum chamber assembly has accompanying documentation regarding leak check and RGA data approval. Confirm that each item in the table passes by typing "Yes" in the appropriate column. If the answer is "No," contact the Responsible Mechanical Engineer and use a different vacuum chamber.

Item	Yes / No	Assembler and Date	Notes
Did RGA data pass for specified vacuum level per LCLS requirements?		[Type your name here] [Place date here]	

Item	Yes / No	Assembler and Date	Notes
Leak check passed at 2x10 ⁻¹⁰ std cc/sec?		[Type your name here] [Place date here]	

8.2 Ladder Cooling Assembly ([LCL5006-005605](#)) Hydrostatic Test



LCL5006-005605 - Ladder Cooling Assembly

- For the Ladder Cooling Assembly ([LCL5006-005605](#)), bend one tube of each circuit at 45 degrees as required.
- Attach and tighten the Swagelok tube fittings (items 6 and 14 of [DSG-000043770](#)) per [Swagelok procedures](#) at the ends of the tubes.
- Then, perform a hydrostatic test of each cooling water supply/return tubes using the following steps:
 - Add water with a water pump to one of the tubes.
 - Connect the hose from the foot pump to one end of the filled tube. Block the other tube end.
 - Use the foot pump to pressurize the tubes to 10 PSI and hold for 10 minutes. NIL is confirmed when there is no drop in pressure for 10 minutes as shown on the pressure gauge.
 - Fill in the data below and repeat the steps for the second tube.
- After doing the hydrostatic test, clean the part, specifically the tubes by blowing them out with dry air, then, bag the stem.

Item	Pressure Drop (PSI)	Tester and Date	Note
Front Tube		[Type your name here] [Place date here]	

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Item	Pressure Drop (PSI)	Tester and Date	Note
Back Tube		[Type your name here] [Place date here]	

Note: The back tube is closest to the mounting plate of the ladder cooling assembly.

8.3 Bellows ([LCL5004-001216](#)) Leak Check and UHV Processing

- Each bellows has been leak checked. Fill in the table below to confirm that the leak test passed using the accompanied documentation and upload the data to the assembly's folder. If the leak test has not passed, contact the Responsible Mechanical Engineer and use a different bellows.

Item	Yes / No	Assembler and Date	Notes
Leak check passed at 2×10^{-10} std cc/sec?		[Type your name here] [Place date here]	
Is the leak check data uploaded to the folder?		[Type your name here] [Place date here]	

- Check one of the following options regarding the UHV processing of the bellows:

- ☐ Baked out and RGA done by vendor.
- ☐ Cleaned and RGA done by Mechanical Fabrication Department (MFD).
- ☐ Need to still be processed.

If the vendor or MFD performed UHV processing on the bellows, add the RGA scan and other relevant documentation in the assembly's folder and fill out the table below. If the bellows still needs to be processed, do so in Section 7.4.

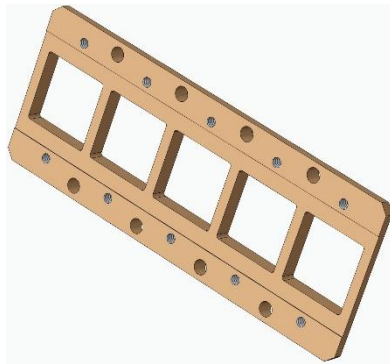
Item	Yes / No	Assembler and Date	Notes
Did RGA data pass for specified vacuum level per LCLS requirements?		[Type your name here] [Place date here]	
Are the relevant files regarding the UHV processing uploaded to the folder?		[Type your name here] [Place date here]	

8.4 Bake Out and RGA the Parts of HE Target Mover Assembly ([DSG-000043946](#))

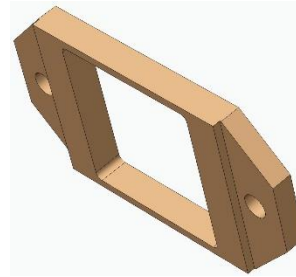
- Bake then RGA the following parts of the HE Target Mover Assembly. If using LCLS bake stations, comply with [SLAC-I-030-050-R0 LCLS Bake Station Procedure](#). Fill out the table below. Add more rows to the table, if needed.

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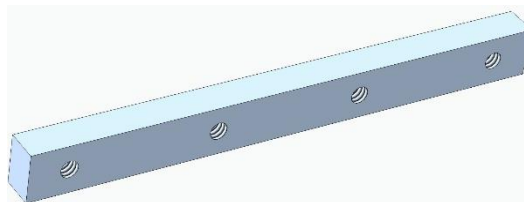
- a. (1) Ladder Cooling Assembly ([LCL5006-005605](#)) from Section 7.2. **Section 7.2 must be completed before bake out.**
 - b. (1) Bellows ([LCL5004-001216](#)) if indicated in Section 7.3.
 - c. (1) Target Base Plate ([DSG-000043818](#))
 - d. (5) Target Frames ([DSG-000043789](#))
 - e. (2) Ladder Nut Blocks ([LCL5010-005627](#))
 - f. Accu-Glass Parts
 - i. (1) 9-Pin Feedthrough Flange (Accu-Glass 100200 item 21)
 - ii. (1) Macor Ceramic D-Connector (Accu-Glass 110900, item 22)
 - iii. (9) Female Contacts (Accu-Glass 100180, item 23)
 - g. (2) RTDs (item 19)
2. Upload the RGA data to this assembly's folder.



DSG-000043818 – Target Base Plate



DSG-000043789 – Target Frame



LCL5010-005627 – Ladder Nut Block

Batch	Items in Batch	Bake Technician, Temperature and Duration	RGA Passed? (Yes / No)	RGA Data Uploaded to Folder? (Yes / No)	Bake Technician and Date Final Sign Off
1	(e.g., - 1 Target Base Plate - 5 Target Frame Plates - 2 Ladder Nut Block)	Technician: Bake Temp: Start Date: End Date:			[Type your name here] [Place date here]
2					[Type your name here] [Place date here]
3					[Type your name here] [Place date here]

8.5 Prepare the RTDs shown in the HE Target Mover Assembly ([DSG-000043946](#))

Prepare the RTDs for installation by attaching the female contacts (Accu-Glass 100180, item 23) to the ends of the wires.

Item	Assembler and Date	Notes
RTD 1	[Type your name here] [Place date here]	
RTD 2	[Type your name here] [Place date here]	

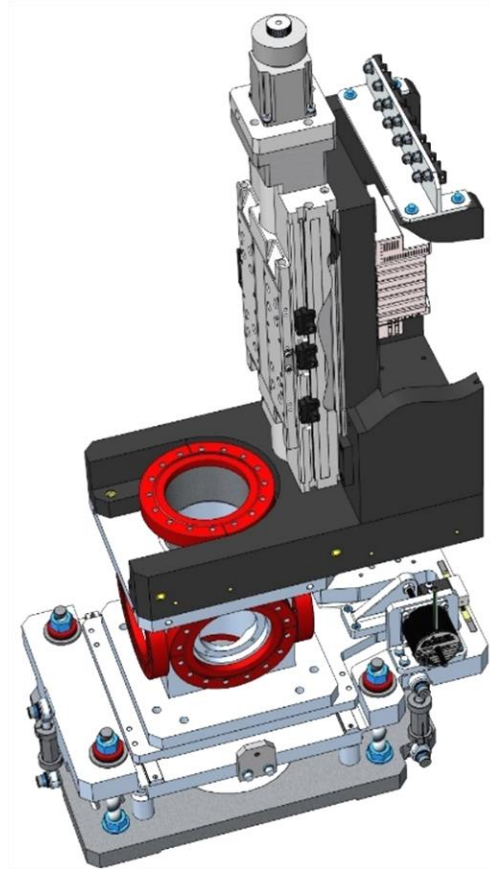


9 HE Wave Front Sensor Core Assy

9.1 Assemble the Chamber to the Stage and Tower

Responsible Person(s): Assembly Technician

If there are any problems or concerns, contact the Responsible Mechanical Engineer.



1. Perform a functional check on the WFS Stage Assembly of the bearing plate (LCL5002-003873, item 1 in [LCL5002-003852](#)) to verify the plate smoothly slides from limit to limit by applying consistent force by hand.
2. Visually inspect the 6-Port Vacuum Chamber ([LCL5004-001164](#)). Check that the table in Section 7.1 is filled out and each item has passed and that the subcontractor documentation for verification of the physical QC assembly data, RGA and bake out data, and vacuum leak check data are uploaded to this assembly's folder. **If there are any problems or concerns, contact the Responsible Mechanical Engineer.**
3. Assemble the 6-Port Vacuum Chamber ([LCL5004-001164](#)) to WFS Stage Assembly ([LCL5002-003852](#)) per [DSG-000043770](#).
 - a. Install chamber to stage with screws (item 10) torqued to at least 240in-lbs.
 - b. Orientate the chamber on the stage so that the nitrogen pinch-off port is at the top of the chamber and the quartz window port is along the X direction (not within the beam path).

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Item	Assembler and Date	Notes
Step 9.1.1: Stage passed functional check	[Type your name here] [Place date here]	
Step 9.1.2: Chamber passed visual inspection	[Type your name here] [Place date here]	
Step 9.1.3: Chamber installed onto stage and in correct orientation	[Type your name here] [Place date here]	

9.2 Assemble the HE WFS Tower Assembly ([DSG-000044220](#))

Responsible Person(s): Assembly Technician

If there are any problems or concerns, contact the Responsible Mechanical Engineer.

1. Perform a functionality test of the Parker Stage ([DSG-000067212](#)).

Item	Assembler and Date	Notes
Brake releases and engages	[Type your name here] [Place date here]	
Stage moves from end to end	[Type your name here] [Place date here]	

2. Assemble the HE WFS Tower Assembly ([DSG-000044220](#)).

- a. Install base plate (LCL5004-003026) to stage tower (LCL5004-004358) with screws (item 10) torqued to at least 520 in-lbs.
- b. Install toe clamps (SL060801-00257, item 22) to stage tower with screws (item 11) torqued to at least 33 in-lbs.
- c. Install Parker stage ([DSG-000067212](#)) to stage tower with screws (item 12) torqued to at least 70 in-lbs.
- d. Install DIN rail ([LCL5004-004420](#)) to stage tower with screws (item 13) torqued to at least 33 in-lbs.
- e. Install tie mounts (SL031000-23119) and the cable mount bracket ([LCL5004-004413](#)) onto stage tower as shown in [DSG-000044220](#).
- f. Do not install the Beckhoff Control modules. These parts will be installed at a later step.
- g. Do not install the Electronics Cover ([LCL5004-004692](#)) nor the Motor Cover ([LCL5010-006088](#)). These covers will be installed at a later step.

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3. Assemble HE WFS Tower Assembly ([DSG-000044220](#)) to top of the chamber ([LCL5004-001164](#)) per [DSG-000043770](#).
 - a. Install chamber to tower base plate with screws (item 10) torqued to at least to 240 in-lbs.
4. Place the Beam Direction labels onto a visible place either on the stage ([LCL5002-003852](#)) or the chamber ([LCL5004-001164](#)).
 - a. Upon installation, the stage tower (LCL5004-004358) faces the South wall (-X direction) in XPP (Hutch 1.3). As such, when looking at the DIN rail of the stage tower, the beam direction is as shown:  **BEAM DIRECTION** 
5. Place labels to the Parker Stage and each wire.

Notes:

1. Torque values of the fasteners were specified in SLAC Engineering Note [LCLSII-2.4-EN-0726-R1](#).
2. This 6-port chamber assembly is the Wave Front Sensor Modular Core assembly sequence.

Item	Assembler and Date	Notes
Step 9.2.2.a: Base Plate to Stage Tower using item 10 in DSG-000044220	[Type your name here] [Place date here]	
Step 9.2.2.b: Toe Clamps to Stage Tower using item 11 in DSG-000044220	[Type your name here] [Place date here]	
Step 9.2.2.c: Parker Stage to Stage Tower using item 12 in DSG-000044220	[Type your name here] [Place date here]	
Step 9.2.2.d: DIN Rail to Stage Tower using item 13 in DSG-000044220	[Type your name here] [Place date here]	
Step 9.2.2.e: Tie Mounts and Cable Mount Bracket onto Stage Tower shown in DSG-000044220	[Type your name here] [Place date here]	
Confirm tower Assembly (DSG-000044220) assembled per step 9.2.2	[Type your name here] [Place date here]	
Step 9.2.3: Tower to Chamber using item 10 in DSG-000043770	[Type your name here] [Place date here]	

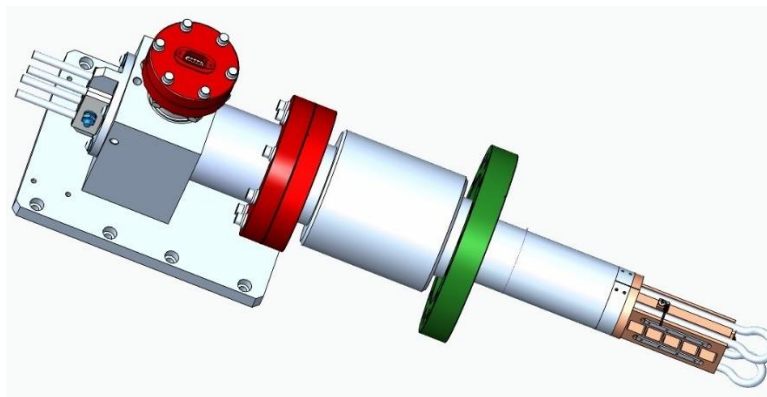
Item	Assembler and Date	Notes
Step 9.2.4: Beam Direction labels are placed	[Type your name here] [Place date here]	Label Location:
Step 9.2.5: Parker Stage labels are placed	[Type your name here] [Place date here]	

10 HE Target Mover Assembly

10.1 Assemble HE Target Mover Assembly

Responsible Person(s): Assembly Technician

If there are any problems or concerns, contact the Responsible Mechanical Engineer.



DSG-000043946 - HE Target Mover Assembly

1. Perform a functionality test of the 2 RTDs. Use a multimeter/ohmmeter to confirm the RTD is working by measuring a resistance of about 110 ohms between the two leads.

Item	Tester and Date	Resistance Value [Ω]	Notes
RTD 1	[Type your name here] [Place date here]		
RTD 2	[Type your name here] [Place date here]		

2. Install the RTDs onto the Ladder Cooling Assembly ([LCL5006-005605](#)) per [DSG-000043946](#) and the [Lake Shore installation sheet](#).

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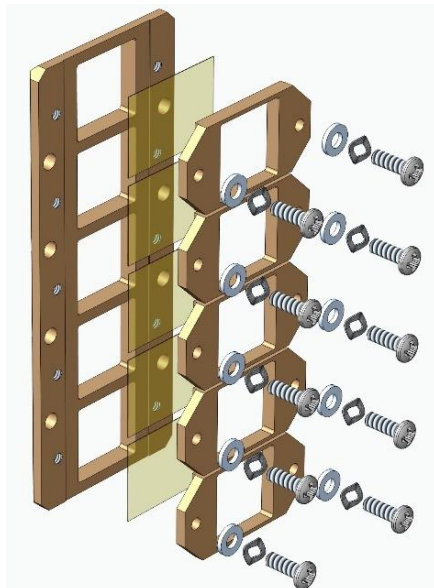
3. Install the Cooling Loop Clamp ([LCL5010-005721](#)) and Clamp Bar ([LCL5010-005817](#)) per [DSG-000043946](#).
4. Install the 9-Pin Feedthrough Flange (Accu-Glass 100200, item 21), Macor Ceramic D-Connector (Accu-Glass 110900, item 22), and Female Contacts (Accu-Glass 100180, item 23). Complete in vacuum wiring of RTDs to 9-pin connector flange per [WD-391-701-08-C00](#).
5. Install bellows module ([LCL5004-001216](#)). Only hand tighten the 4.5 CF flange bolts.

Item	Assembler and Date	Notes
Step 10.1.2: Side RTD Installed	[Type your name here] [Place date here]	
Step 10.1.2: Bottom RTD Installed	[Type your name here] [Place date here]	
Step 10.1.3: Cooling Loop Clamp and Clamp Bar installed	[Type your name here] [Place date here]	
Step 10.1.4: Accu-Glass parts installed	[Type your name here] [Place date here]	
Step 10.1.4: In-vacuum wiring of RTDs completed	[Type your name here] [Place date here]	
Step 10.1.5: Bellows module installed	[Type your name here] [Place date here]	

10.2 Assemble the HE Target Assembly

Responsible Person(s): Assembly Technician

If there are any problems or concerns, contact the Responsible Mechanical Engineer.



DSG-000043817 - HE Target Assembly

1. Assemble the HE Target Assembly ([DSG-000043817](#)) and document the targets below.
2. Take a picture of each target onto the base plate (DSG-000043818) before clamping on the frame plate (DSG-000043789). Upload the pictures into the assembly folder.

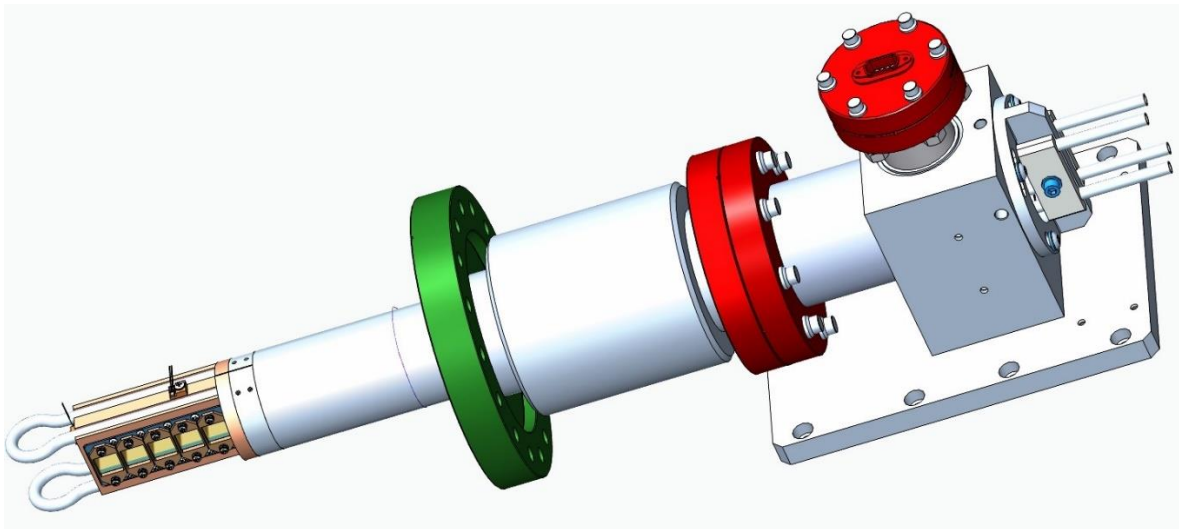
Target	Target Grating ID	Material and Orientation	Assembler and Date	Scientist Sign Off and Date
1 (Bottom)			[Type your name here] [Place date here]	[Type your name here] [Place date here]
2			[Type your name here] [Place date here]	[Type your name here] [Place date here]
3			[Type your name here] [Place date here]	[Type your name here] [Place date here]
4			[Type your name here] [Place date here]	[Type your name here] [Place date here]

Target	Target Grating ID	Material and Orientation	Assembler and Date	Scientist Sign Off and Date
			[Place date here]	[Place date here]
5 (Top)			[Type your name here] [Place date here]	[Type your name here] [Place date here]

10.3 Complete the assembly of the HE Target Mover Assembly

Responsible Person(s): Assembly Technician

If there are any problems or concerns, contact the Responsible Mechanical Engineer.



DSG-000043946 - HE Target Mover Assembly

1. Install the HE Target Assembly ([DSG-000043817](#)) onto the Ladder Cooling Assembly ([LCL5006-005605](#)) per [DSG-000043946](#).

Item	Assembler and Date	Notes
HE Target Assembly Installed	[Type your name here] [Place date here]	

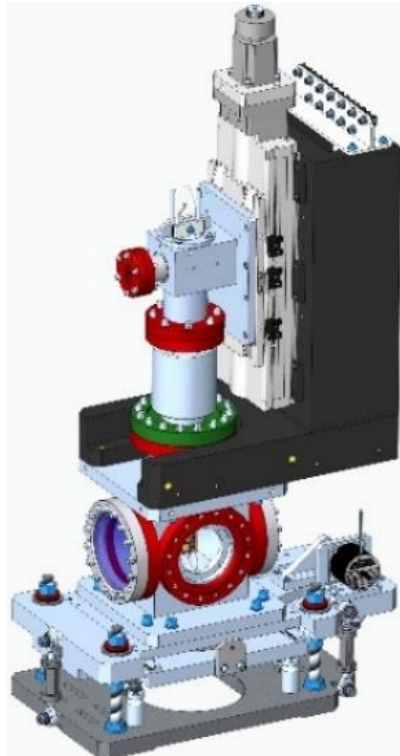
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11 HE Wavefront Sensor Assembly

Responsible Person(s): Assembly Technician

If there are any problems or concerns, contact the Responsible Mechanical Engineer.

Assemble the Core assembly from Section 9, with HE Target Mover Assembly from Section 10.3.



DSG-000043770-Mover Installation

1. Install the Target Mover Assembly ([DSG-000043946](#)) onto the Parker Stage ([DSG-000067212](#)) with a 6.0 CF gasket in place. Use mounting shims as needed to flatly secure the mover to the Parker stage. Torque fasteners to at least 70 in-lbs as specified in SLAC Engineering Note [LCLSII-2.4-EN-0726-R1](#).
2. Install the Target Mover ([DSG-000043946](#)) to the chamber using 6.0 CF flange hardware.
3. Tighten the 4.50 CF flange hardware.
4. Perform a functionality test of the Parker Stage ([DSG-000067212](#)) with the Target Mover Assembly attached.

Item	Assembler and Date	Notes
Step 11.1: Target Mover onto Parker Stage	[Type your name here] [Place date here]	



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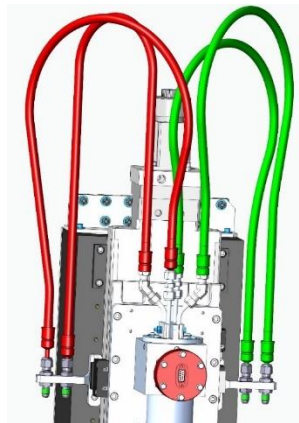
Item	Assembler and Date	Notes
Step 11.2: Bellows onto Chamber	[Type your name here] [Place date here]	
Step 11.3: Tighten 4.50 CF flange hardware	[Type your name here] [Place date here]	
Step 11.4: Parker Stage moved full length of travel with Target mover	[Type your name here] [Place date here]	

5. Install the second viewport – 6.0 Quartz Viewport (MDC 450024, item 18) per [DSG-000043770](#).

Item	Assembler and Date	Notes
Step 11.5: Viewport Installed	[Type your name here] [Place date here]	

11.1 Install remaining water-cooling fixtures

Install water fitting bracket ([DSG-000001357](#), item 1), bulkhead unions (item 7), and braided hoses (item 8) of [DSG-000043770](#) as shown in the drawing.

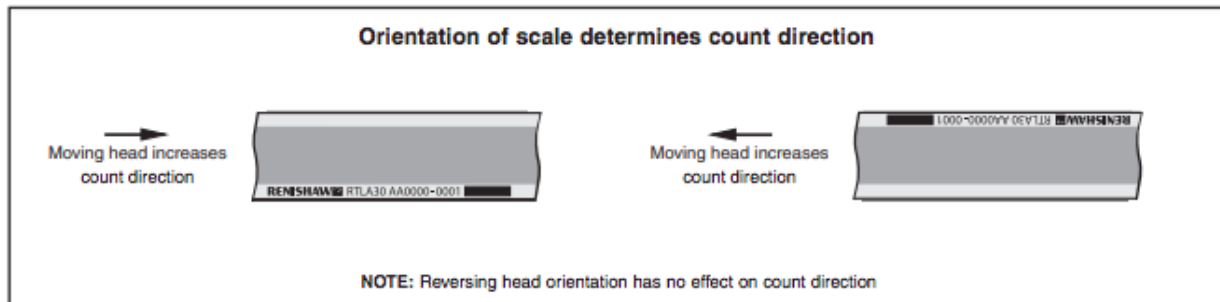


Item	Assembler and Date	Notes
Water Fitter Bracket	[Type your name here] [Place date here]	
Bulkhead Unions	[Type your name here] [Place date here]	

Item	Assembler and Date	Notes
Braided Hoses	[Type your name here] [Place date here]	

11.2 Install motor and encoder onto WFS Stage Assembly ([LCL5002-003852](#))

1. Perform a functionality test of the motor (item 12). Install the motor (item 12), as shown in [LCL5002-003852](#). Note the serial number of the motor in the table below. Create and add the respective label to the motor.
2. Install the encoder (item 17), encoder scale (item 15), and other Renishaw items (13, 14, 16, and 38) as shown in [LCL5002-003852](#). The encoder scale is oriented to have positive counts in the +Z direction (i.e., beam direction). See Appendix 20.1 for encoder mounting information.
 - a. Check for proper alignment of the encoder using the ECS alignment tool per the [Renishaw ADTa Encoder Testing SOP](#). Encoder light shall be blue (preferred) or green (still okay) across full travel as verification of proper alignment and signal.
 - b. Note the serial number of the encoder in the table below.
 - c. Note the direction of the positive (+) travel (i.e., which way does the assembly move when the encoder counts are increasing?) in terms of the LCLS coordinate system.



Item	Assembler and Date	Notes
Step 11.2.1: Motor installed with label	[Type your name here] [Place date here]	Motor serial number:
Step 11.2.2: Encoder installed and aligned	[Type your name here] [Place date here]	Encoder serial number: Direction of (+) travel:
Confirm LCL5002-003852 WFS Stage is fully assembled per drawing	[Type your name here] [Place date here]	

11.3 Install Beckhoff Modules onto DIN Rail

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1. Confirm Beckhoff modules from bag labeled “R-000036801” are correctly assembled per [WD-391-701-08](#).
2. Attach Beckhoff modules onto DIN rail.

Item	Assembler and Date	Notes
Step 11.3.1: Confirmed Beckhoff modules are correctly assembled	[Type your name here] [Place date here]	
Step 11.3.2: Attached Beckhoff modules onto DIN Rail	[Type your name here] [Place date here]	

12 Wiring and Functional Test

12.1 Wiring

Responsible Person(s): Assembly Technician

If there are any problems or concerns, contact the Responsible Mechanical Engineer.

1. Create and add labels to the necessary wires and components.
2. Wire Beckhoff control modules per [WD-391-701-08](#).
3. Complete the [pre-power check procedure](#).

Item	Assembler and Date	Notes
Step 12.1.1: All labels are placed	[Type your name here] [Place date here]	
Step 12.1.2: Beckhoff control modules wired	[Type your name here] [Place date here]	
Step 12.1.3: Pre-power check completed	[Type your name here] [Place date here]	

12.2 Device Testing

Responsible Person(s): ECS Engineer

If there are any problems or concerns, contact the Responsible Mechanical Engineer.

1. Complete the table below.



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Item	Action / What to Check
Z-Axis Stepper Motor on Stage	Verify the motor works by sliding bearing plate (LCL5002-003873) from end to end and finishing at starting position.
Z-Axis Encoder on Stage	Verify data is being read out correctly.
Y-Axis Brake on Parker Stage	Release the break
Y-Axis Motor on Parker Stage	Verify the motor works by moving stage from end to end and finishing at starting position.
Y-Axis Encoder on Parker Stage	Encoder light to be blue (preferred) or green (still okay) across full travel to verify proper alignment and signal. Verify data is being read out correctly.
Y-Axis Limit Switches on Parker Stage	Verify limit switches report the necessary signal when activated and not activated.
RTDs	Verify that each RTD works by outputting a room temperature value.

Item	Passed? Yes/No	Assembler and Date	Witness Sign Off and Date	Notes
Z-Axis Stepper Motor on Stage		[Type your name here] [Place date here]	[Type your name here] [Place date here]	
Z-Axis Encoder on Stage		[Type your name here] [Place date here]	[Type your name here] [Place date here]	
Y-Axis Brake on Parker Stage		[Type your name here] [Place date here]	[Type your name here] [Place date here]	
Y-Axis Motor on Parker Stage		[Type your name here] [Place date here]	[Type your name here] [Place date here]	

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Item	Passed? Yes/No	Assembler and Date	Witness Sign Off and Date	Notes
		[Place date here]	[Place date here]	
Y-Axis Encoder on Parker Stage		[Type your name here] [Place date here]	[Type your name here] [Place date here]	
Y-Axis Limit Switches on Parker Stage		[Type your name here] [Place date here]	[Type your name here] [Place date here]	
Side RTD		[Type your name here] [Place date here]	[Type your name here] [Place date here]	
Bottom RTD		[Type your name here] [Place date here]	[Type your name here] [Place date here]	

Note: If an item does not pass, check the remaining items. Document the items that failed and the part serial number in the notes of the table. Remove and replace the part. Document the new serial number, add the necessary labels, and run the tests again. Contact the Responsible Mechanical Engineer and place the defective part aside to give to the engineer.

12.3 Calibration of Z-Axis stage

Responsible Person(s): ECS Engineer and Mechanical Engineer or Assembly Technician

3. Move stage to up beam limit (+Z), 2-3 mm before the hard stop, and record encoder value.
4. Move stage to down beam limit (-Z), 2-3 mm before the hard stop, and record encoder value.
5. Find the median between the +Z and -Z hard stop encoder values. Record the encoder value as the "Zero Position" as this will be the middle of travel. Move the stage to the zero position.
6. Power down the equipment when the steps are completed and/or when leaving the assembly unattended.
- 7.

Target	Aligner and Date	Encoder Value	Position (mm)	Witness Sign Off and Date	Notes
+Z Hard Stop	[Type your name here] [Place date here]			[Type your name here] [Place date here]	
-Z Hard Stop	[Type your name here] [Place date here]			[Type your name here] [Place date here]	
Zero Position	[Type your name here] [Place date here]			[Type your name here] [Place date here]	

13 HE Target Mover Alignment

Responsible Person(s): Mechanical Engineer or Assembly Technician and ECS Engineer (if requested)

13.1 Set Parker stage limits

1. Install the calibration disc with the pinhole onto the chamber and steer the alignment laser through the pinhole.
2. Move the Y stage until the bottom of the stalk clips the alignment laser in half.
3. Move +15 mm from this position and set the upper limit switch less than 5 mm above this point.
4. Move -86 mm from the upper limit switch position and set the lower limit switch less than 5 mm below this point.

Limit Switch	Aligner and Date	Witness and Date (if applicable)	Notes
Upper	[Type your name here] [Place date here]	[Type your name here] [Place date here]	
Lower	[Type your name here] [Place date here]	[Type your name here] [Place date here]	

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13.2 Calibration of diamond target center

Move the Parker stage to the center of the bottom target (target 1) diamond target, referencing the adjacent fastener as the target center. Fill out the table below for each subsequent target by moving the Parker stage appropriately to each target center.

Target	Aligner and Date	Encoder Value	Witness and Date (if applicable)	Notes
1	[Type your name here] [Place date here]		[Type your name here] [Place date here]	
2	[Type your name here] [Place date here]		[Type your name here] [Place date here]	
3	[Type your name here] [Place date here]		[Type your name here] [Place date here]	
4	[Type your name here] [Place date here]		[Type your name here] [Place date here]	
5	[Type your name here] [Place date here]		[Type your name here] [Place date here]	

14 Fiducialization

Responsible Person(s): SLAC Metrology Group and ECS Engineer (if requested)

1. Working with the SLAC Metrology Group, calibrate the target pitch and yaw with respect to the tooling ball sockets. This is accomplished by reflecting a laser off of the target surface, measuring the angle between the incident and exit beam.

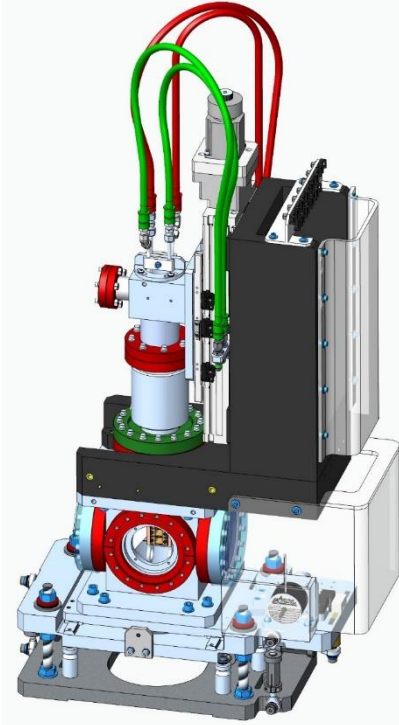
Report Name	[Device Name must be in the Report Name]
Report Location	[Teamcenter or Assembly Folder?]
Alignment Date	[Place date here]
Aligner Sign Off	[Type your name here]
Witness(es) (if applicable)	[Type your name(s) here]

15 Install Safety Covers

Responsible Person(s): Assembly Technician

If there are any problems or concerns, contact the Responsible Mechanical Engineer.

Install respective covers over electric connections ([LCL5004-004692](#), item 5), motor ([LCL5010-006088](#), item 7), and viewport flanges (Torr Scientific VPZ100-COVER, item 9) per [DSG-000043770](#).



Item	Assembler and Date	Notes
Electronics Cover	[Type your name here] [Place date here]	
Motor Cover	[Type your name here] [Place date here]	
Viewport Cover (2)	[Type your name here] [Place date here]	

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16 Place Labels

Responsible Person(s): Assembly Technician

If there are any problems or concerns, contact the Responsible Mechanical Engineer.

Place part number and asset name labels on the assembly in a visible place. Note the label positions in the table below.

Item	Assembler and Date	Notes
Part Number Label	[Type your name here] [Place date here]	
Asset Name Label	[Type your name here] [Place date here]	

17 Final Leak Check, RGA, and Nitrogen Back Fill

Responsible Person(s): Assembly Technician

If there are any problems or concerns, contact the Responsible Mechanical Engineer.

1. Install Z-axis pinch off flanges onto the chamber.
2. Perform a final leak check in accordance with [SLAC-I-120-204-000-00](#) Helium. Fill in the table below.
3. Complete a final RGA scan in accordance with [SLAC-I-030-714-001-09-R1](#) RGA Acceptance Criteria for RFI. Fill in the table below.
4. Back fill the chamber with nitrogen through the pinch off flanges installed in step 1.

Leak Checker and Date	Leak Rate	Passed? (Yes / No)	Notes
[Type your name here] [Place date here]			
RGA Scan Approver and Date	Uploaded Scan to Folder? (Yes / No)	Passed? (Yes / No)	
[Type your name here] [Place date here]			

18 Ready for Installation (RFI) Final Approval

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If there are any problems or concerns that prevents you from providing final approval of this completed assembly traveler, contact the Responsible Mechanical Engineer.

Approver Name	Role	Signature and Date	Notes
	Quality Assurance Representative		
Leinani Roylo	Common Components Lead Engineer		
Diling Zhu	XPP HE Lead Instrument Scientist		
Ruben Curiel	Lead Assembly Technician		
Divya Kameswaran	Common Components Lead ECS Engineer		
Rebecca Armenta	XPP HE Lead Instrument Engineer		



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19 Appendix A: Additional Information

19.1 Encoder Installation

Readhead mounting/installation

Mounting brackets

The bracket must have a flat mounting surface, enable conformance to the installation tolerances, allow adjustment of the rideheight of the readhead, and be sufficiently stiff to prevent deflection of the readhead during operation.

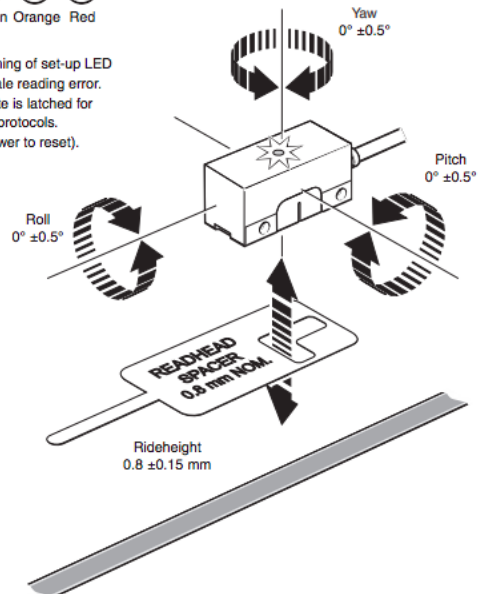
Readhead set-up

Ensure that the scale, readhead optical window and mounting face are clean and free from obstructions. To set nominal rideheight for RTLA-S installations, use the Blue 0.8 mm readhead spacer to set rideheight. Adjust the readhead to maximize the signal strength along the full axis of travel to achieve a Green or Blue LED.

Readhead set-up LED and DRIVE-CLIQ interface set-up STATUS LED status

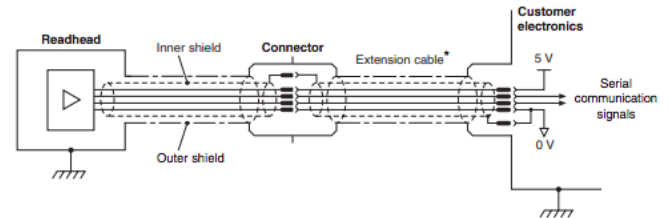


NOTE: Flashing of set-up LED indicates scale reading error. Flashing state is latched for some serial protocols. (Remove power to reset).



Electrical connections

RESOLUTE grounding and shielding



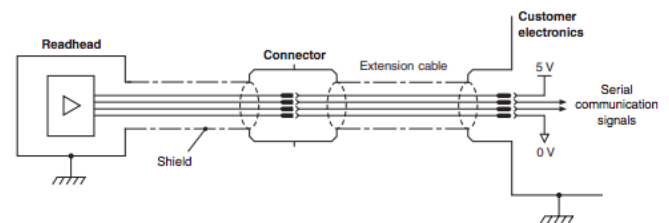
IMPORTANT: The outer shield should be connected to the machine earth (Field ground). The inner shield[†] should be connected to 0 V at customer electronics only. Care should be taken to ensure that the inner and outer shields are insulated from each other. If the inner and outer shields are connected together, this will cause a short between 0 V and earth, which could cause electrical noise issues.

IMPORTANT: If the flying lead variant is used or the connector is modified or replaced, the customer must ensure both 0 V cores (White and Green) are connected to 0 V.

^{*}For Siemens DRIVE-CLIQ please refer to Siemens DRIVE-CLIQ specifications for maximum cable length.

[†]For UHV there is no inner shield to be connected.

RESOLUTE grounding and shielding - Yaskawa protocol only



IMPORTANT: The shield should be connected to the machine earth (Field ground).

IMPORTANT: If the connector is modified or replaced, the customer must ensure both 0 V cores (White and Green) are connected to 0 V.



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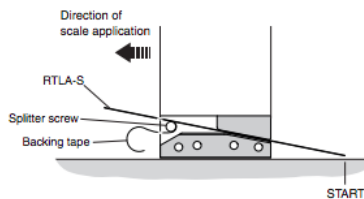
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Scale application

The scale applicator A-9589-0095 is designed specifically for use with RESOLUTE readheads and RTLA-S scale.

- ① Allow scale to acclimatize to installation environment prior to installation.
- ② Mark out the 'START' and 'FINISH' points for the scale on the axis substrate – ensure that there is room for the optional end covers if required (see 'RTLA-S installation drawing').
- ③ Thoroughly clean and degrease the substrate using recommended solvents see 'Storage and handling'. Allow substrate to dry before applying scale.
- ④ Mount the appropriate scale applicator to the readhead mounting bracket using M3 screws. Place the shim supplied with the readhead between the applicator and substrate to set the nominal height.
NOTE: Scale applicator can be mounted either way round to enable easiest orientation for scale installation.
- ⑤ Move axis to 'START' of travel.
- ⑥ Begin to remove the backing paper from the scale and insert scale into the applicator up to the 'START' point (as shown). Ensure backing tape is routed under the splitter screw.
- ⑦ Apply finger pressure to the scale at the 'START' point, using a clean lint-free cloth, to ensure scale end adheres well to the substrate.
- ⑧ Slowly and smoothly move the applicator through the entire axis of travel, ensuring the backing paper is pulled manually from the scale and does not catch under the applicator.



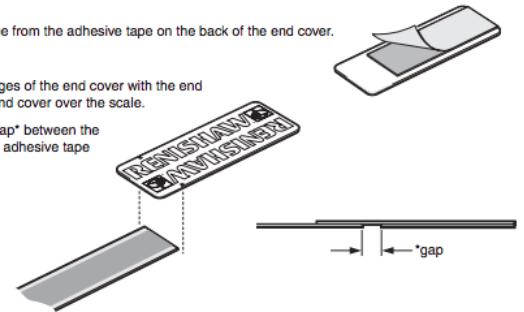
- ⑨ Remove applicator and, if necessary, adhere the remaining scale manually. Apply firm finger pressure via a clean lint-free cloth along the length of the scale after application to ensure complete adhesion.
- ⑩ Clean scale using Renishaw scale cleaning wipes (A-9523-4040) or a clean, dry, lint-free cloth.
- ⑪ Fit end covers.
- ⑫ Allow 24 hours for complete adhesion of scale before fitting datum clamps.

End covers

The end cover kit A-9585-0035 is designed to be used with RTLA-S scale to provide protection for exposed scale ends.

NOTE: End covers are optional and can be fitted before or after readhead installation.

- ① Remove the backing tape from the adhesive tape on the back of the end cover.
- ② Align markers on the edges of the end cover with the end of the scale and place end cover over the scale.
NOTE: There will be a gap* between the end of the scale and the adhesive tape on the end cover.

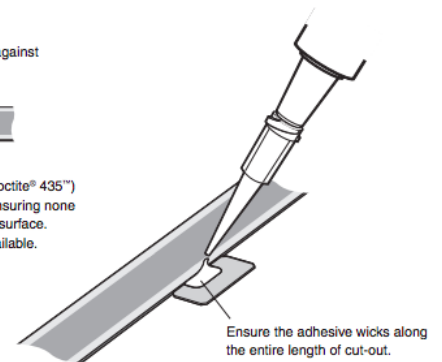


Datum clamp (A-9585-0028)

The datum clamp fixes the RTLA-S scale rigidly to the substrate at the location chosen.

The metrology of the system may be compromised if the datum clamp is not used. It can be positioned anywhere along the axis depending upon the customers' requirements.

- ① Place the datum clamp with cut-out against the scale at the chosen location.
- ② Place a small amount of adhesive (Loctite® 435™) in the cut-out on the datum clamp, ensuring none of the adhesive wicks onto the scale surface. Dispensing tips P-TL50-0209 are available.



Output signals

Function	Signal ^a	Wire colour
Power	5 V	Brown
	0 V	White
		Green
Serial communications	MA+	Violet
	MA–	Yellow
	SLO+	Grey
	SLO–	Pink
Shield	Inner ^a	Inner shield
	Outer	Outer shield

19.2 Additional Manufacturing Notes

Name	Notes	Date